FORTYGUARD

Heat Intelligence Report

Heatmap Metadata

Location:	33.44839999948182, -112.074
Date and Time:	2024-07-15 12:00
Temperature:	35.0°C (95.0°F)

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Map



Figure 1: Overview Area

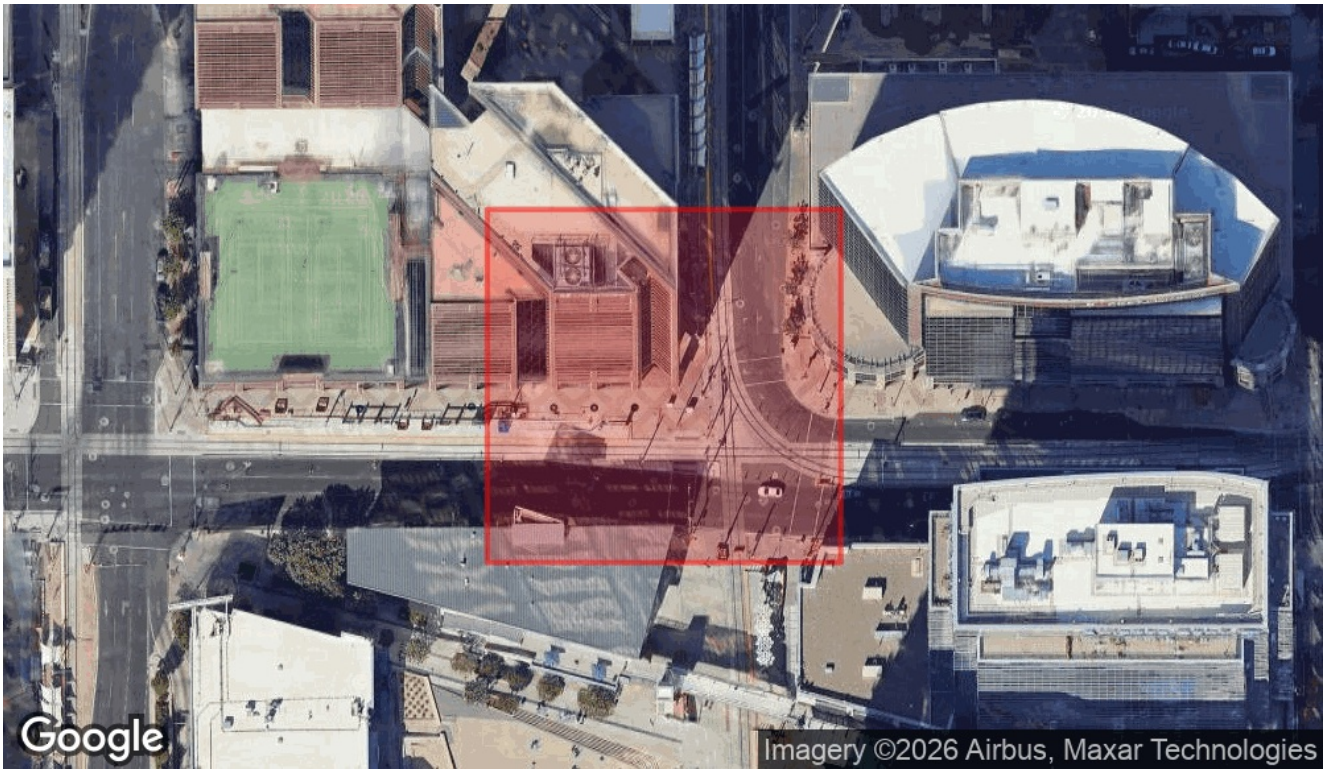


Figure 2: Tile Area

Environmental Factors Analysis

1. Air Quality & Atmospheric Opacity

Typical AQI Range & Classification

- Phoenix, Arizona in mid-July typically experiences AQI values ranging from 50–120, falling in the EPA categories of **Good to Unhealthy for Sensitive Groups**.
- Ozone (O_3) is the dominant concern during summer, frequently pushing AQI into the Moderate (51–100) or Unhealthy for Sensitive Groups (101–150) range during afternoon hours. PM10 spikes occur during dust storm (haboob) events, which are characteristic of the North American Monsoon season onset in July.

Primary Pollutants & Sources

- **Ozone (O_3):** Formed photochemically from NO_x and volatile organic compounds (VOCs) emitted by vehicle traffic (~4.5 million metro population), industrial operations, and fuel combustion. Phoenix's intense solar radiation and high temperatures accelerate ozone formation significantly.
- **PM10:** Elevated by windblown desert dust, construction activity, unpaved roads, and haboob events. During haboobs, PM10 can exceed 500 $\mu g/m^3$, pushing AQI well into Hazardous territory.
- **PM2.5:** Moderate concern from vehicular emissions, wildfire smoke transport (from California, New Mexico, or northern Arizona fires), and secondary aerosol formation.
- **NO_2 and CO:** Primarily from vehicular emissions; generally within acceptable limits but contribute to ozone precursor chemistry.

Influence on Local Temperature

- Elevated particulate matter acts as a dual radiative agent: fine aerosols scatter incoming shortwave radiation (slight surface cooling effect) while also absorbing and re-emitting longwave radiation (warming effect). In Phoenix's dry atmosphere, the net effect of dust events is typically a slight reduction in daytime peak temperature (1–3°F) due to solar dimming, but enhanced nighttime warmth from trapped thermal radiation.
- Smog and haze reduce direct solar transmittance but increase diffuse radiation, partially offsetting the cooling effect on total absorbed energy.

Heat-Ozone Feedback Loop

- At 95°F (and typical Phoenix afternoons exceeding 105°F at the surface), ozone production rates are significantly amplified. Ground-level ozone formation roughly doubles for every 18°F increase above ~75°F baseline temperatures.
- Higher temperatures also increase biogenic VOC emissions from sparse desert vegetation and increase evaporative emissions from fuels, further feeding ozone precursor loads.
- Stagnant high-pressure conditions common in July trap pollutants in the lower boundary layer, concentrating both ozone and particulates.

Sources:

- EPA AQI standards and pollutant breakpoint tables
- Maricopa County Air Quality Department monitoring data patterns
- NOAA haboob climatology for the Desert Southwest
- Peer-reviewed literature on photochemical ozone formation kinetics

2. Climate Classification & Historical Weather

Köppen Climate Classification

- Phoenix is classified as **BWh — Hot Desert Climate** under the Köppen system. This classification is characterized by extremely hot summers, mild winters, very low annual precipitation (typically ~8.0 inches/year), and large diurnal temperature ranges. Aridity is the defining characteristic, with potential evapotranspiration far exceeding precipitation year-round.

Historical Climate Normals (July 15)

- **Typical daily high (1991–2020 normals):** ~106°F
- **Typical daily low:** ~84°F
- **Typical daily mean:** ~95°F
- The provided average temperature of **95.0°F is exactly in line with the climatological daily mean** for mid-July in Phoenix. This value is consistent with the normal climate profile and does not represent an anomalous reading. Note, however, that 95°F at noon (12:00 hours) would be below the typical instantaneous temperature at that hour, which is usually around 100–104°F, suggesting this may be a daily average or that conditions were moderated by cloud cover or early monsoon moisture.

Extreme Temperature Records

- **All-time record high:** 122°F (June 26, 1990, at Phoenix Sky Harbor International Airport).
- **July record high:** 121°F (July 28, 1995).
- In 2023, Phoenix experienced a record-breaking streak of 31 consecutive days at or above 110°F (July).
- Record daily mean temperatures have been set repeatedly in recent years, with 24-hour means exceeding 100°F on the hottest days.

Long-Term Climate Profile & Urban Heat Behavior

- The BWh classification means the urban environment operates with virtually no evaporative cooling from natural vegetation or soil moisture for most of the year. This amplifies the effectiveness of anthropogenic heat sources and impervious surface thermal storage in driving UHI.
- Clear skies prevail ~85% of the time in June-July (pre-monsoon), maximizing solar radiation receipt and surface heating.
- The arrival of the North American Monsoon (typically mid-July) introduces intermittent cloud cover, increased humidity, and convective rainfall, which can temporarily moderate heat but also increase heat index values.

Sources:

- NOAA 1991–2020 Climate Normals for Phoenix Sky Harbor International Airport (KPHX)
- Köppen climate classification system
- NOAA extreme weather records database
- National Weather Service Phoenix office historical data

3. Climate Trends & Projections

Observed Temperature Trends

- Phoenix has warmed at an accelerated rate compared to global averages. The observed warming trend is approximately **0.5–0.7°F per decade** since 1970, with nighttime minimum temperatures warming even faster (~1.0°F per decade) due to the compounding effects of UHI expansion and regional climate change.
- The number of days per year exceeding 110°F has increased from an average of ~12 days/year in the 1970s to ~20+ days/year in recent decades.
- Heatwave frequency and duration have increased markedly. Consecutive-day heat events (≥5 days above 110°F) have become more common since 2000.
- Relative humidity trends are mixed: June (pre-monsoon) has become slightly drier, while monsoon-season moisture delivery shows high interannual variability with no strong secular trend.

Urban Expansion & Land-Use Change

- The Phoenix metropolitan area has expanded dramatically: the population grew from ~2.0 million (1990) to ~4.9 million (2023), with corresponding conversion of desert and agricultural land to impervious urban surfaces.
- Estimated: Urban land area in the Phoenix metro has roughly tripled since 1980, replacing native Sonoran Desert (albedo ~0.30–0.35) with asphalt and concrete (albedo ~0.10–0.25), directly amplifying heat absorption.
- Loss of irrigated agricultural land (historically common in the Salt River Valley) has removed an evaporative cooling source, contributing an estimated 1–2°F of additional warming in some formerly agricultural zones.

Climate Change Projections

- **Near-term (2030s):** Under RCP 4.5/SSP2-4.5 scenarios, mean summer temperatures in the Phoenix area are projected to increase by an additional 1.5–2.5°F relative to the 2000–2020 baseline. Days above 110°F could average 25–35 per year.
- **Mid-century (2050s):** Under moderate emission scenarios, July mean temperatures could increase 3–5°F, with extreme heat days (≥115°F) becoming significantly more frequent. Under high-emission scenarios (SSP5-8.5), increases of 5–7°F in summer mean temperatures are projected.
- Monsoon precipitation is projected to become more episodic — fewer but more intense events — which could increase flash flood risk while providing less consistent evaporative cooling relief.

Cooling Degree Days (CDD) Trend

- Phoenix already has one of the highest annual CDD totals in the U.S., approximately **4,500–5,000 CDD (base 65°F)**. This has been increasing at roughly 100–150 CDD per decade.
- Rising CDD translates directly to increased electricity demand for air conditioning, which feeds back into anthropogenic heat emissions and grid stress.

Sources:

- NOAA U.S. Climate Divisional Dataset
- NASA GISS Surface Temperature Analysis (GISTEMP)
- National Climate Assessment (NCA4 and NCA5 findings for the Southwest)
- IPCC AR6 regional projections for the North American Southwest
- U.S. Census Bureau population data

4. Soil Moisture & Permeability

Soil Types & Permeability

- The Phoenix basin sits in the Salt River Valley, with predominant soil types being **arid sandy loam, caliche (calcium carbonate hardpan), and alluvial desert soils**.
- Native desert soils have moderate to high permeability in the sandy upper horizons but are often underlain by caliche layers at 1–4 feet depth, which create impermeable barriers that prevent deep infiltration.
- Soil taxonomy: primarily Aridisols and Entisols, characterized by low organic matter (<1%), minimal biological activity, and poor water retention.

Seasonal Soil Moisture Conditions

- By mid-July, soil moisture is typically at or near its annual minimum. Pre-monsoon conditions (late June through early July) are the driest period, and July 15 sits at the transition point when monsoon moisture may or may not have arrived.
- Estimated: Volumetric soil moisture content in non-irrigated areas is typically 2–5% in the upper 12 inches — essentially desiccated.
- Irrigated urban landscapes (lawns, parks, golf courses) represent isolated pockets of elevated soil moisture and provide localized evaporative cooling.

Heat Absorption & Evaporative Cooling

- Dry desert soils have low thermal conductivity (~0.3 W/m·K) but moderate heat capacity, meaning they heat intensely at the surface while insulating subsurface layers. Surface temperatures of bare desert soil can reach 150–165°F under direct July sun.
- With negligible soil moisture, virtually no latent heat flux (evaporative cooling) occurs from natural ground surfaces. Nearly all absorbed solar energy is partitioned into sensible heat flux, directly warming the air — this is the fundamental driver of extreme daytime temperatures in arid environments.
- The Bowen ratio (sensible to latent heat flux) in the pre-monsoon Phoenix area typically exceeds 10:1, compared to ~1:1 in humid environments.

Urbanization Impacts

- Estimated: 55–65% of the Phoenix urban core is covered by impervious surfaces (buildings, roads, parking lots), virtually eliminating natural soil permeability and infiltration.
- Soil sealing replaces natural desert soil (which, despite low moisture, has some thermal buffering) with high-thermal-mass materials that store more heat and release it at night.
- Stormwater infrastructure routes the rare heavy monsoon rains away rapidly, preventing infiltration that could recharge soil moisture.

Groundwater Influence

- The groundwater table in central Phoenix is generally 100–300 feet below the surface due to extensive historical pumping, rendering it irrelevant to surface soil moisture or evaporative cooling at the ground level.

Sources:

- USDA Natural Resources Conservation Service (NRCS) Soil Survey for Maricopa County
- USDA soil taxonomy classification system
- Peer-reviewed research on arid soil thermal properties and Bowen ratio partitioning
- Arizona Department of Water Resources groundwater data

5. Heat Retention Capacity of Surfaces

Dominant Surface Materials

- **Asphalt:** Comprises roadways, parking lots, and driveways — estimated 30–40% of developed land area. Dark asphalt has an albedo of ~0.05–0.15 (new to weathered).
- **Concrete:** Sidewalks, commercial surfaces, building foundations — albedo ~0.20–0.40 depending on age and composition.
- **Bare desert soil and gravel landscaping:** Common in residential xeriscape — albedo ~0.25–0.35. Lower thermal mass than asphalt/concrete but still significant heat storage.
- **Building rooftops:** Mix of flat commercial roofs (often light-colored or gravel-topped, albedo ~0.20–0.60) and residential tile roofs (terra cotta, albedo ~0.30–0.40).
- **Vegetation:** Limited to irrigated urban parks, street trees (primarily Palo Verde, mesquite, and non-native species), and residential landscaping. Green fraction in central Phoenix is estimated at only 10–15%.

Surface Temperature Differentials Under Peak Solar Load

- At solar noon in July, surface temperature differentials are dramatic:
  - **Black asphalt:** 150–170°F surface temperature
  - **Concrete:** 130–150°F
  - **Bare desert soil:** 140–160°F
  - **Irrigated grass:** 95–105°F
  - **Tree-shaded surfaces:** 85–100°F
  - **Cool/reflective pavement:** 110–125°F
- The difference between asphalt and irrigated grass can exceed **60–70°F** in surface temperature, illustrating the enormous cooling potential of vegetation and moisture.

Thermal Admittance & Albedo

- **Asphalt:** High thermal admittance (~1,500–2,000 J/m²·s^½·K), meaning it absorbs heat rapidly and stores large quantities. Combined with low albedo, it is the single most problematic surface for UHI.
- **Concrete:** Moderate thermal admittance (~1,200–1,700 J/m²·s^½·K), somewhat better albedo mitigates daytime peak but still stores substantial heat.
- **Desert gravel/soil:** Lower thermal admittance (~600–1,000 J/m²·s^½·K), releases stored heat more quickly after sunset.

UHI Contribution

- Surface material composition is the primary driver of Phoenix's UHI, which has been measured at **5–12°F** above

surrounding desert at night. Daytime UHI in Phoenix is typically smaller (2–5°F) because the surrounding desert itself is already extremely hot.
- The nocturnal UHI is dominated by heat release from high-thermal-mass materials: asphalt and concrete continue radiating stored heat well into the night, maintaining elevated air temperatures.

Cooling Infrastructure

- The City of Phoenix has piloted cool pavement programs (CoolSeal and similar reflective coatings), reporting surface temperature reductions of 10–12°F on treated surfaces.
- Cool roof incentive programs exist, particularly for commercial buildings. Arizona State University research parks have deployed high-albedo roofing as demonstration projects.
- Urban tree canopy goals have been established (targeting 25% canopy cover from the current ~10%), but implementation is slow due to water constraints and species survival challenges.

Sources:

- EPA "Reducing Urban Heat Islands: Compendium of Strategies"
- City of Phoenix Cool Pavement Pilot Program monitoring reports
- ASU Global Institute of Sustainability research on urban heat in Phoenix
- Peer-reviewed literature on surface material thermal properties and urban albedo

6. Humidity Levels

Typical Relative Humidity

- July 15 falls at the onset of the North American Monsoon. Humidity conditions are highly dependent on whether the monsoon has actively established:
  - **Pre-monsoon (dry):** Relative humidity at noon can be as low as 5–15%.
  - **Active monsoon:** Noon relative humidity rises to 20–40%, with early morning values of 40–60%.
- Typical range for July 15 at noon: **15–30% RH**, reflecting the transitional monsoon period.

Dew Point Temperature

- Pre-monsoon dew points: 30–45°F (extremely dry, comfortable).
- Active monsoon dew points: 55–68°F (noticeably more humid, occasionally uncomfortable).
- Estimated dew point for this date/time: **45–60°F**, depending on monsoon moisture advection. The monsoon onset is typically declared when the Phoenix dew point averages above 55°F for three consecutive days.
- Even at the higher end of monsoon dew points, Phoenix rarely approaches the dangerous threshold of >75°F dew point that characterizes extreme heat stress in humid climates.

Heat Index Implications

- At 95°F with 20% RH, the heat index is approximately **93–95°F** — essentially equal to or slightly below the actual temperature, as low humidity allows effective evaporative cooling of the skin.
- If monsoon moisture elevates humidity to 40% at 95°F, heat index rises to approximately **99–101°F**, a meaningful increase in perceived heat.
- At Phoenix's more typical July peak of 110°F with 15% RH, the heat index is ~105–108°F. With monsoon moisture (30% RH at 110°F), heat index jumps to ~120°F+ — a dangerous combination.

Seasonal Humidity Patterns

- Phoenix has a bimodal humidity regime: extremely dry from April through late June (monthly mean RH ~10–20%), then a sharp increase during the monsoon (July–September, mean RH ~25–40%).
- October through March sees moderate dryness with occasional Pacific storm moisture.
- The monsoon humidity increase is critical because it coincides with the hottest temperatures, creating compound heat stress events that are more dangerous than dry heat alone.

Nocturnal Cooling Suppression

- Increased monsoon moisture significantly suppresses nighttime radiative cooling. Water vapor absorbs outgoing longwave radiation, acting as a greenhouse blanket. On humid monsoon nights, minimum temperatures may remain above 90°F — compared to 80–85°F on dry nights — creating consecutive "tropical nights" that prevent physiological recovery from daytime heat stress.

Sources:

- NOAA Climate Normals for Phoenix Sky Harbor (humidity and dew point data)
- National Weather Service Heat Index calculation tables
- Peer-reviewed literature on North American Monsoon climatology
- NWS Phoenix monsoon onset criteria

7. Solar Radiation Intensity

Global Horizontal Irradiance (GHI)

- Phoenix receives some of the highest solar radiation in North America. On July 15 under clear-sky conditions, GHI at solar noon is approximately **950–1,050 W/m²**.
- Daily total GHI in mid-July averages approximately **7.5–8.5 kWh/m²/day** under clear conditions, making it one of the most solar-intense locations in the continental U.S.
- If monsoon cloud buildup occurs (common in afternoons), GHI can drop to 300–600 W/m² during convective activity but typically remains at peak values through the noon hour.

Solar Elevation Angle

- At 33.44°N latitude on July 15 at solar noon (approximately 12:30 PM local time due to Phoenix's position within the Mountain Standard Time zone), the solar elevation angle is approximately **80**° (nearly directly overhead).
- This near-vertical sun angle minimizes shadow lengths, maximizes radiation intensity per unit area of horizontal surface, and ensures that urban canyon floors, roads, and parking lots receive nearly direct-beam radiation with minimal building shading.
- The high solar elevation also means that vertical building facades receive relatively less direct radiation than horizontal surfaces, concentrating heating on rooftops, roads, and ground-level surfaces.

Peak Sunlight Hours & Thermal Lag

- Sunrise is approximately 5:30 AM; sunset approximately 7:40 PM, yielding ~14.2 hours of daylight.
- Peak solar intensity occurs between 10:00 AM and 2:00 PM.
- Maximum air temperature typically lags solar noon by 2–4 hours, peaking around 3:00–5:00 PM. This thermal lag is driven by the time required for surface materials to absorb, store, and re-radiate heat to the air boundary layer.
- At noon (the given time), the temperature of 95°F is still rising toward the afternoon peak, which typically reaches 105–110°F.

Terrain & Urban Modification of Solar Exposure

- Phoenix's flat valley topography offers no significant terrain shading. The surrounding mountain ranges (South Mountain, Camelback, McDowell Mountains) shade only small localized areas at very low sun angles (sunrise/sunset).
- Urban structures provide limited mutual shading due to Phoenix's predominantly low-rise, spread-out urban form (1–3 story buildings dominate the landscape). The central business district (near the given coordinates) has some mid-rise to high-rise buildings that create intermittent canyon shading.
- Typical cloud cover in July mornings is minimal (<10%); afternoon convective clouds build from the monsoon but rarely develop before 1:00–2:00 PM.

Differential Surface Absorption

- Under ~1,000 W/m² of solar radiation, asphalt absorbs approximately 850–950 W/m², while high-albedo cool pavement reflects 350–500 W/m² and absorbs 500–650 W/m². Irrigated grass absorbs similar total radiation but partitions ~50–70% into latent heat (evapotranspiration) rather than sensible heat.

Sources:

- NREL National Solar Radiation Database (NSRDB)
- NOAA Solar Calculator for solar elevation angles
- NASA Surface Meteorology and Solar Energy (SSE) dataset
- Peer-reviewed literature on surface energy balance partitioning in arid urban environments

8. Nighttime Cooling Rates

Diurnal Temperature Range (DTR)

- The typical DTR in Phoenix during July is approximately **20–27°F**. With a normal high of ~106°F and low of ~84°F, the DTR is about 22°F.
- This is a relatively compressed DTR compared to the surrounding open desert, where DTR can reach 30–40°F, reflecting the urban heat island's suppression of nighttime cooling.
- During monsoon humidity surges, DTR contracts further to 15–20°F as water vapor traps outgoing longwave radiation.

Surface Thermal Mass & Urban Geometry Effects

- High-thermal-mass surfaces (asphalt, concrete, masonry) absorb substantial heat during the 14+ hours of daylight and release it slowly via longwave radiation through the night. Asphalt surfaces may still be 100–110°F at midnight.
- Phoenix's urban geometry, while generally low-rise, includes wide arterial roads (often 6–8 lanes) that act as significant heat reservoirs. Commercial districts with large parking lots (sky view factor near 1.0) radiate more efficiently than narrow residential streets but have more absolute thermal mass to shed.
- In the downtown area near the given coordinates, buildings with height-to-width ratios creating street canyons trap longwave radiation through multiple reflections between building facades and road surfaces, reducing the effective sky view factor and slowing radiative cooling.

Nighttime Cooling Classification

- Phoenix exhibits **weak nighttime cooling** relative to the surrounding environment, particularly in the urban core. This is one of the most significant health hazards of Phoenix summers.
- Nighttime lows above 90°F ("super tropical nights") have occurred with increasing frequency. In 2023, Phoenix recorded nights that failed to cool below 95°F.
- Consecutive hot nights prevent physiological thermoregulatory recovery, significantly increasing heat-related morbidity and mortality. The human body requires temperatures below ~80°F for effective sleep-time cooling; Phoenix regularly fails to reach this threshold for extended periods in July.

Factors Suppressing Nighttime Cooling

- **Thermal mass:** The single largest factor — billions of square feet of heat-storing surfaces continue releasing energy through the night.
- **Monsoon humidity:** Water vapor's strong longwave absorption bands trap outgoing radiation.
- **Anthropogenic heat:** Air conditioning systems reject enormous quantities of waste heat — Estimated: metropolitan Phoenix HVAC systems release ~5,000–10,000 MW of thermal energy during peak cooling periods, contributing measurably to nocturnal air temperatures.
- **Minimal vegetation:** Low evapotranspiration rates mean no significant nighttime latent cooling.
- **Cloud cover:** Monsoon clouds, when present overnight, completely block longwave radiation escape.

Urban vs. Rural Nighttime Differential

- The nocturnal UHI in Phoenix has been measured at **7–12°F**, with the urban core (near Sky Harbor Airport and the given coordinates) being warmest and outlying desert areas coolest.
- Irrigated agricultural areas (increasingly rare) and undeveloped desert preserve land show the largest cooling differentials relative to the urban core.

Sources:

- NOAA climate normals for diurnal temperature range
- ASU urban climate research (Dr. David Sailor, Dr. Ariane Middel, and colleagues)

- Peer-reviewed literature on nocturnal UHI in desert cities
- NWS Phoenix heat records

9. Thermal Comfort Index

Applicable Thermal Comfort Metric

- For Phoenix's arid environment, the **Wet Bulb Globe Temperature (WBGT)** and **Heat Index (HI)** are both commonly applied, but the most comprehensive metric is the **Universal Thermal Climate Index (UTCI)**, which integrates air temperature, radiation, wind, and humidity into a single physiologically meaningful value.
- However, the **Mean Radiant Temperature (MRT)**-driven **PET (Physiological Equivalent Temperature)** is particularly relevant for Phoenix because in arid high-radiation environments, radiant heat load from surfaces and direct sun is often the dominant thermal stressor — more so than humidity. MRT in direct sun in Phoenix can exceed 160°F (71°C) at noon in July.
- For practical application and public communication, the **NWS Heat Index** remains the standard, though it underestimates danger in direct-sun, high-radiation desert conditions.

Combined Factor Analysis

- **Air temperature:** 95°F (at noon, rising to ~106°F by afternoon peak).
- **Humidity:** ~20–30% RH (monsoon transitional).
- **Wind speed:** Typical July midday winds are 5–10 mph, providing minimal convective cooling relief.
- **Mean Radiant Temperature (MRT):** Estimated at 140–165°F in direct sun due to high solar elevation, intense GHI, and hot surrounding surfaces radiating in all directions. In shade, MRT drops dramatically to 100–115°F.

Estimated Comfort Classification

- **Heat Index:** At 95°F and 25% RH, HI ≈ 95°F — classified as "Caution" (fatigue possible with prolonged exposure).
- **WBGT:** Estimated at ~82–86°F, falling in the "High" to "Extreme" risk category for physical exertion (OSHA/military guidelines recommend limiting strenuous activity above WBGT of 85°F).
- **UTCI (in sun):** Estimated at ~110–120°F (43–49°C), classified as **"Very Strong Heat Stress"** to **"Extreme Heat Stress"**.
- **UTCI (in shade):** Estimated at ~95–102°F (35–39°C), classified as **"Strong Heat Stress."**
- Note: The discrepancy between Heat Index ("Caution") and UTCI ("Very Strong/Extreme") highlights the critical importance of radiant heat in desert environments that standard HI calculations miss.

Heat Stress Risks for Vulnerable Populations

- Maricopa County (Phoenix metro) consistently leads the nation in heat-associated deaths, with ~300+ heat-related fatalities in 2023.
- **Unhoused populations:** The most vulnerable group, with direct prolonged sun exposure and limited access to cooling.
- **Outdoor workers:** Construction, landscaping, and agricultural workers face WBGT levels requiring mandatory rest/hydration breaks.
- **Elderly and chronically ill:** Particularly those without functioning AC or unable to afford electricity costs, at risk of indoor heat stroke.
- **Children:** Higher surface-area-to-body-mass ratio and proximity to superheated ground surfaces (a child at 3 feet height experiences temperatures 5–10°F hotter than an adult at 5.5 feet).

Local Mitigation Measures

- **Cooling centers:** Over 60 designated cooling centers operated by the City of Phoenix and Maricopa County during heat season.
- **Heat relief network:** Nonprofit and faith-based organizations provide water stations, shade, and cooling access.
- **Shaded infrastructure:** Phoenix has installed shade structures at transit stops and along pedestrian corridors. ASU research (the "Shade for Phoenix" initiative) identified areas needing priority shade investment.
- **Misting stations:** Deployed at outdoor venues and some public transit locations, providing ~10–15°F perceived temperature reduction via evaporative cooling.

- **NWS Excessive Heat Warnings:** Routinely issued for extended periods in July, with public outreach campaigns ("Beat the Heat").

Sources:

- NWS Heat Index tables and WBGT calculation methodology
- UTCI assessment scale (International Society of Biometeorology)
- Maricopa County Department of Public Health heat surveillance reports
- ASU Extreme Heat Research (Nature's Cooling Systems project)
- OSHA heat illness prevention guidelines

10. Seasonal Variability of Urban Heat Islands

UHI Intensity by Season

- **Summer (June–August):** Nocturnal UHI intensity is at its annual maximum, typically **7–12°F**. Daytime UHI is moderate (2–5°F) because the surrounding desert is also extremely hot. The large thermal mass of urban surfaces combined with anthropogenic heat rejection (AC waste heat) drives the nighttime maximum.
- **Fall (September–November):** UHI intensity moderates to 4–8°F as temperatures decrease and AC loads diminish. Reduced solar angle lowers surface heat storage.
- **Winter (December–February):** UHI intensity is weakest, typically **3–6°F**. Shorter days reduce thermal storage. Some heating-related anthropogenic heat exists but is far less than summer AC loads. The UHI can actually be beneficial in winter, reducing heating demand.
- **Spring (March–May):** UHI rebuilds as temperatures and solar intensity increase, reaching 5–9°F by May. This is the period of most rapid UHI intensification as desert vegetation dries and urban surfaces begin maximum heat storage.

Driving Factors

- **Foliage cycles:** Phoenix's limited deciduous vegetation means foliage changes have minimal impact on UHI compared to temperate cities. Evergreen desert-adapted species (Palo Verde, mesquite) provide consistent but sparse shade year-round. Irrigated landscaping with deciduous non-natives provides slightly more summer shade.
- **Monsoon effects:** The July–September monsoon introduces intermittent cloud cover, rainfall, and increased humidity. Rainfall events provide brief but dramatic cooling (surface temperatures can drop 30–40°F during and immediately after storms). However, post-storm humidity inhibits nocturnal radiative cooling, partially offsetting the benefit.
- **Seasonal energy use:** Summer AC electricity consumption in Phoenix can exceed 25–30 GWh/day for the metro area. Every unit of electrical energy used for cooling is ultimately rejected as heat to the outdoor environment (plus the thermodynamic inefficiency), creating a positive feedback loop where AC waste heat warms the outdoor air, requiring more AC.
- **Snow/ice cover:** Not applicable. Phoenix experiences no snow or ice cover that would alter albedo dynamics.

Urban-Rural Temperature Differentials Across Seasons

- Summer: Urban core 7–12°F warmer than outlying desert at night; 2–5°F during daytime.
- Winter: Urban core 3–6°F warmer at night; negligible to slightly warmer during daytime.
- The most extreme urban-rural contrast occurs during calm, clear summer nights when radiative cooling is maximized in the desert but suppressed in the city.

Seasonal Adaptation Measures

- **Summer-specific:** Cooling center activation (May–October), adjusted outdoor work schedules (many construction projects shift to pre-dawn hours), park and recreation schedule modifications, expanded transit shade.
- **Year-round infrastructure:** Cool pavement programs (most effective in summer), tree planting initiatives (most planting occurs in cooler months for establishment), building energy code improvements requiring higher-efficiency HVAC and reflective roofing.
- **Planned:** The City of Phoenix adopted a Heat Action Plan and Climate Action Plan with targets for increased canopy cover (25% goal), expanded cool surfaces, and heat equity mapping to prioritize vulnerable neighborhoods.

Greatest Compound Heat Risk

- **Late June through mid-July (pre-monsoon to monsoon onset) presents the greatest compound heat risk.** This period combines:
  - Maximum solar intensity and day length
  - Peak air temperatures (highest daily means of the year)
  - Dry soils with zero evaporative cooling
  - Maximum UHI intensity
  - Potential for monsoon moisture arrival that raises humidity without yet providing cooling rainfall — creating humid-hot compound events
  - Peak AC energy demand stressing the electrical grid (risk of outages that would be catastrophic for public health)
- July 15 — the date of this analysis — sits precisely within this highest-risk window. The combination of extreme heat, UHI amplification, potential monsoon humidity surges, and vulnerable population exposure makes this period the deadliest of the year, as confirmed by Maricopa County mortality data showing July as the peak month for heat-related deaths.

Sources:

- ASU Urban Climate Research Center UHI monitoring network data
- City of Phoenix Climate Action Plan and Heat Action Plan
- Maricopa County heat mortality surveillance data
- EPA UHI characterization studies for desert cities
- Peer-reviewed literature on seasonal UHI dynamics in arid metropolitan areas

Geographic Analysis

1. General Location

Identification

- **City:** Phoenix, Arizona, United States
- **Geographic classification:** Urban core of a major metropolitan area situated in an inland arid desert basin (Salt River Valley)
- **Köppen climate zone:** BWh — Hot Desert Climate, characterized by extremely high summer temperatures, very low annual precipitation (approximately 8 inches/year), intense solar radiation, and large diurnal temperature swings

Historical Temperature Extremes & Patterns

- **All-time record high:** 122°F (June 26, 1990)
- **Typical July highs:** The average daily high in Phoenix for mid-July is approximately 106–108°F, with average daily lows around 84–87°F. The overall daily mean temperature for July 15 is approximately 95–97°F.
- **Comparison to input temperature:** The provided average temperature of 95.0°F at noon is consistent with the climate normal daily mean for this date but is significantly **below** the expected peak daytime high of ~106–108°F. At solar noon, ambient temperatures are typically rising toward their daily peak (usually reached around 3:00–5:00 PM local time), so 95°F at noon is plausible as temperatures are still climbing. However, this reading could also reflect a slightly cooler-than-average day, localized shade, or a well-irrigated microsite.
- **Notable heat patterns:** Phoenix has experienced an intensifying UHI effect over decades, with nighttime lows rising faster than daytime highs. The city regularly experiences multi-day heat waves exceeding 110°F. In 2023, Phoenix recorded a historic 31 consecutive days at or above 110°F. Heat-related mortality is a significant public health concern, with Maricopa County recording over 600 heat-associated deaths in 2023.

Sources:

- NOAA National Weather Service — Phoenix climate normals and records
- Köppen climate classification system
- Maricopa County Department of Public Health — heat surveillance reports

2. Terrain & Elevation

Elevation & Terrain Type

- **Elevation:** Approximately 1,085 feet above sea level at downtown Phoenix
- **Terrain type:** Phoenix sits in the broad, relatively flat Salt River Valley (Valley of the Sun), a low-elevation desert basin. The valley floor is generally flat with gentle slopes, surrounded by isolated mountain ranges and buttes (Camelback Mountain at ~2,704 ft, South Mountain at ~2,690 ft, the McDowell Mountains, White Tank Mountains, and Sierra Estrella).

Terrain Influence on Heat Dynamics

- **Basin heat retention:** The valley basin configuration inhibits effective heat dissipation. Surrounding mountains partially block regional wind flow, reducing ventilation and trapping heated air within the valley. This exacerbates both daytime heating and nocturnal heat retention.
- **Nocturnal inversions:** On calm, clear nights, radiative cooling of elevated terrain can produce katabatic (downslope) drainage flows, but these are typically weak and insufficient to flush heat from the urbanized valley floor. Temperature inversions can form, trapping warm air and pollutants near the surface.
- **Solar radiation:** The flat, open terrain allows unobstructed solar radiation across vast surface areas. At this latitude (33.44°N), the solar elevation angle on July 15 at solar noon is approximately 80°, meaning nearly vertical insolation and maximum energy input per unit area. Slope orientation is largely irrelevant across the flat valley floor but matters on the surrounding mountain flanks — south-facing slopes receive more prolonged heating.
- **Thermal mass of desert floor:** The valley's bare soil and rock surfaces absorb and re-radiate significant heat, contributing to elevated surface temperatures that can exceed 160°F on exposed asphalt.

Sources:

- USGS elevation data (National Elevation Dataset)
- NOAA solar position calculations
- Arizona State University urban climate research

3. Proximity to Water Bodies

Nearest Significant Water Bodies

- **Salt River:** Flows through/adjacent to the southern edge of downtown Phoenix (approximately 2–3 miles south of the coordinates). However, the Salt River is largely dry through the metro area for most of the year due to upstream damming (Roosevelt Dam, Horse Mesa Dam, etc.). Tempe Town Lake, an artificial impoundment on the Salt River approximately 7 miles east-southeast, provides a small localized water body (~220 acres).
- **Gila River:** Approximately 20 miles to the south; also largely dry through the metro area.
- **Canals:** The Salt River Project (SRP) and Central Arizona Project (CAP) canal systems thread through the metro area. The Arizona Canal passes approximately 5 miles north of downtown. These narrow water channels provide negligible regional thermal moderation.
- **Nearest large reservoir:** Lake Pleasant is approximately 35 miles northwest; Saguaro Lake is approximately 35 miles northeast.

Thermal Moderating Effect

- **Minimal buffering:** Phoenix receives effectively **no meaningful thermal moderation from water bodies.** The absence of a nearby ocean, large lake, or perennial river means there is no maritime or lacustrine influence to dampen temperature extremes. This is a primary reason Phoenix experiences some of the highest urban temperatures in North America.
- **Humidity:** Ambient humidity is generally very low (typical July relative humidity is 20–35% in early afternoon). During the North American Monsoon (typically mid-July through September), moisture surges from the Gulf of California can temporarily raise dewpoints to 55–65°F, slightly reducing the diurnal temperature range but increasing heat stress (heat index).
- **Irrigation effects:** Widespread landscape irrigation and agricultural use in the surrounding metro area contribute some localized evaporative cooling and increased humidity, partially offsetting the arid baseline. Estimated urban "oasis effect" from irrigation can reduce temperatures by 2–5°F in well-watered residential areas compared to unirrigated desert.
- **Prevailing winds:** Dominant summer winds are from the south to southwest, bringing continental desert air rather than maritime moisture, except during monsoon surges.

Sources:

- Salt River Project infrastructure data
- NOAA — Phoenix monsoon climatology
- Arizona State University — urban oasis effect studies (Brazel et al.)

4. Green Spaces & Vegetation

Type and Extent of Vegetation

- **Urban parks:** Key green spaces near downtown include Encanto Park (~222 acres, approximately 2 miles northwest), Papago Park (~1,500 acres, approximately 6 miles east), and South Mountain Park/Preserve (~16,000 acres, approximately 7 miles south — one of the largest municipal parks in the U.S., though largely undeveloped desert).
- **Street trees and landscaping:** Downtown Phoenix has relatively sparse street tree coverage. Many residential areas feature xeriscape (low-water desert landscaping) or irrigated turf, creating a patchwork of microclimates.
- **Surrounding land:** The broader metro is ringed by Sonoran Desert vegetation (saguaro, palo verde, creosote bush, mesquite) in undeveloped areas and mountain preserves, with agricultural lands (cotton, citrus, alfalfa) historically prominent in outlying areas, though rapidly being converted to development.

Canopy Coverage

- **Estimated tree canopy:** Phoenix has one of the lowest urban tree canopy coverages among major U.S. cities. City-wide canopy coverage is estimated at approximately 9–10%. In the urban core near these coordinates, canopy coverage is likely 5–7% or lower.
- **Vegetation type:** Mix of drought-adapted native species (palo verde, mesquite — deciduous or semi-deciduous, providing moderate summer shade) and non-native species (palms, which provide minimal shade; ornamental trees on irrigated properties). Evergreen species are less dominant than in wetter climates.

Greening Initiatives & Cooling Contribution

- **Tree and Shade Master Plan:** The City of Phoenix adopted a Tree and Shade Master Plan (2010, updated) with goals to increase canopy coverage to 25% by 2030. Progress has been slow due to water constraints, extreme heat stress on new plantings, and rapid development.
- **Cool Corridors:** The city has invested in shade structures at transit stops and along pedestrian corridors, particularly in the urban core.
- **Cooling contribution:** Studies in Phoenix indicate that well-irrigated parks with mature tree canopy can be 5–10°F cooler (ambient air temperature) than surrounding built areas. Surface temperature reductions under tree canopy can exceed 20–30°F compared to exposed pavement. However, the limited extent of green cover means the overall city-scale cooling contribution is modest.

Sources:

- City of Phoenix Tree and Shade Master Plan
- American Forests — urban tree canopy assessments
- Arizona State University — urban ecology and heat mitigation research

5. Land Cover Type

Dominant Surface Materials

- **Downtown core (near coordinates):** Predominantly asphalt roadways and parking lots, concrete sidewalks and structures, commercial/institutional building rooftops (flat, often dark-membrane or built-up roofing), and limited vegetation. Large surface parking lots are particularly prevalent in Phoenix's downtown compared to denser cities.
- **Surrounding residential areas:** Mix of stucco-clad structures with tile or shingle roofs, concrete block walls, decomposed granite yards (xeriscape), irrigated turf lawns, and asphalt streets.
- **Bare soil/desert:** Exposed decomposed granite and caliche soils are common in vacant lots and undeveloped parcels, as well as desert preserves at the metro fringe.

Thermal Properties

- **Asphalt:** Albedo of approximately 0.05–0.15 (very dark, highly absorptive). Surface temperatures on asphalt in direct

July sun routinely reach 150–170°F. Asphalt has high thermal mass and re-radiates stored heat well into the night, significantly contributing to elevated nocturnal temperatures.

- **Concrete:** Albedo of approximately 0.20–0.40. Somewhat more reflective than asphalt but still a significant heat absorber with high thermal mass.
- **Desert soil (decomposed granite):** Albedo of approximately 0.20–0.35. Heats rapidly but has lower thermal mass than asphalt; cools faster at night.
- **Irrigated vegetation/turf:** Albedo of approximately 0.20–0.25, but active evapotranspiration keeps surface temperatures dramatically lower (often 40–60°F cooler than adjacent asphalt surfaces).

Cool Surface Strategies

- **Cool-roof programs:** The City of Phoenix building code encourages high-albedo roofing materials for new commercial construction. Some large commercial buildings and warehouses use white or reflective membrane roofs (albedo ~0.60–0.80).
- **Cool-pavement pilots:** Phoenix has been a national leader in cool-pavement testing. Beginning in 2020, the city partnered with Arizona State University to apply reflective sealcoat treatments (e.g., CoolSeal by GuardTop) to neighborhood streets. Pilot results showed surface temperature reductions of approximately 10–12°F on treated pavement, though some residents reported increased perceived radiant heat at pedestrian level due to reflected solar radiation.
- **Parking lot regulations:** Some newer developments are required to include shade structures or trees covering a percentage of parking lot area.

Sources:

- Arizona State University — Global Institute of Sustainability and Innovation, cool pavement studies
- City of Phoenix — cool roof and shade ordinances
- EPA Heat Island Effect compendium — surface albedo and thermal property data

6. Building Density & Height

Building Density Characterization

- **Downtown Phoenix (at coordinates):** The urban core features a modest cluster of mid-rise to high-rise buildings centered roughly within a 1-mile radius of the intersection of Central Avenue and Washington Street. Building density is **moderate by national standards** — significantly less dense than cities like New York, Chicago, or San Francisco. Large surface parking lots, wide streets, and low-rise structures intersperse between taller buildings, creating a relatively open urban fabric.
- **Surrounding areas:** Rapid transition to low-rise development (1–2 story) within 2–3 miles of the core. The vast majority of the Phoenix metro consists of single-story residential and low-rise commercial/retail strip development, resulting in an extremely horizontally expansive urban footprint (metro area exceeds 14,000 square miles).

Building Heights

- **Tallest structures:** Chase Tower (483 ft), CenturyLink Tower (407 ft), and several buildings in the 200–350 ft range. Approximately 15–20 buildings exceed 200 feet.
- **Typical heights in core:** 4–15 stories (50–200 ft) for most commercial and government buildings.
- **UHI contribution from vertical massing:** The relatively low and dispersed building stock means Phoenix's UHI effect is driven more by **horizontal extent of impervious surfaces and thermal mass** than by canyon-effect radiation trapping seen in denser cities. However, the sheer spatial scale of the metro area (over 500 square miles of developed land) creates one of the most intense and extensive UHIs in the U.S., with nocturnal UHI intensity estimated at 7–12°F compared to surrounding desert.

Common Materials & Anthropogenic Heat

- **Roofing:** Flat commercial roofs with built-up membrane (often dark), residential tile roofs (terra cotta, concrete tile — moderate albedo ~0.25–0.40), and some reflective-coated roofs on newer construction.
- **Facades:** Stucco and concrete masonry unit (CMU) block are ubiquitous in Phoenix construction. Glass curtain walls on newer high-rises. Stucco typically has moderate albedo (~0.30–0.50) depending on color.
- **Anthropogenic heat sources:** Air conditioning is near-universal and runs intensively from May through October. HVAC systems reject enormous quantities of waste heat to the outdoor environment. Estimated metro-area AC

electricity consumption during peak summer days exceeds 10,000 MW of electrical demand (with roughly 3x that in waste heat rejected outdoors). Vehicle traffic on the extensive freeway system (I-10, I-17, Loop 101, Loop 202) generates significant waste heat. Industrial heat sources are dispersed across the metro but include manufacturing, data centers (a rapidly growing sector), and power generation.

Sources:

- Arizona State University — Urban Climate Research Center (UHI monitoring data)
- Emporis/CTBUH — building height databases
- Arizona Public Service (APS) and Salt River Project (SRP) — peak electricity demand data

7. Shadow Coverage

Solar Angle Calculation

- **Date/Time:** July 15, 2024, at 12:00 hours (assumed local time, MST — Arizona does not observe daylight saving time). Solar noon in Phoenix on this date occurs at approximately 12:34 PM MST, so 12:00 PM is very close to solar noon.
- **Solar elevation angle:** At solar noon on July 15 at 33.44°N latitude, the solar declination is approximately +21.5°. The solar elevation angle is calculated as: 90° − |33.44° − 21.5°| ≈ 90° − 11.94° ≈ **78.1°**. The sun is nearly directly overhead, casting very short shadows.
- **Shadow length:** At a solar elevation of ~78°, a 100-foot building casts a shadow only about 21 feet long (100 / tan(78°) ≈ 21.3 ft). This means shadows are confined to the immediate base of structures, providing minimal pedestrian shading at this time of day.

Shadow Coverage Assessment

- **Building shadows:** Due to the near-vertical solar angle, high-rise buildings provide negligible shadow coverage on adjacent streets and open areas at midday. Shadow benefits are significant only in narrow east-west oriented gaps between closely spaced buildings. Shadow coverage from buildings increases substantially in early morning and late afternoon when solar elevation drops below 45°.
- **Tree shadows:** Mature trees (30–40 ft canopy height) cast shadows of only ~7–9 feet in radius at midday. Given the sparse canopy coverage (~5–7% in the core), tree shade is available only in isolated pockets. Palo verde and mesquite trees, while common, are often small to medium-sized with relatively open, filtered canopy.
- **Shade structures:** The city has installed engineered shade structures at bus stops, light rail stations (Valley Metro), and some pedestrian areas. These provide critical but spatially limited relief. The Roosevelt Row and downtown arts district have some shade canopies.

Temperature Impact of Shading

- **Surface temperature reduction:** Shade from buildings or dense tree canopy can reduce surface temperatures by 20–40°F compared to directly sunlit pavement in Phoenix's summer conditions.
- **Ambient air temperature reduction:** Shade reduces pedestrian-level air temperature by approximately 5–9°F compared to adjacent sunlit areas, with more significant reductions in Mean Radiant Temperature (MRT), which is the primary driver of thermal comfort.
- **Permanent vs. seasonal shade:** Building shade is year-round. Most shade trees in Phoenix are semi-deciduous or evergreen desert species that retain foliage in summer (the critical period), making seasonal leaf loss less of a concern than in temperate climates. However, some ornamental deciduous species lose leaves in winter, providing beneficial passive solar heating during cool months.

Sources:

- NOAA Solar Calculator — solar position algorithms
- Arizona State University — outdoor thermal comfort and shade studies (Middel et al.)
- City of Phoenix Cool Corridors program documentation

8. Urban Geometry & Street Canyon Effects

Street Canyon Characteristics

- **Downtown core:** Phoenix's downtown has a rectilinear grid street layout, oriented approximately N-S and E-W. Major streets are wide — Central Avenue and Washington/Jefferson Streets are approximately 80–100 feet curb-to-curb with additional sidewalk width. Including setbacks, right-of-way widths can reach 100–130 feet.
- **Height-to-Width (H/W) ratio:** With typical downtown building heights of 50–200 feet and street widths of 80–130 feet, the H/W ratio in the core ranges from approximately **0.4 to 1.5**, with most streets closer to the lower end. This creates relatively shallow street canyons compared to dense cities like Manhattan (H/W of 2–5). Only a few blocks with the tallest buildings on narrow side streets approach H/W ratios of 1.5–2.0.
- **Longwave radiation trapping:** At H/W ratios below 1.0, canyon radiative trapping is relatively modest. Surfaces can "see" a large portion of the sky and re-radiate longwave energy efficiently at night. In the few deeper canyons (H/W > 1.0), multiple reflections and absorption of longwave radiation between building facades slow nocturnal cooling. However, this effect is less significant in Phoenix's UHI compared to the sheer thermal mass of horizontally extensive impervious surfaces.

Wind Corridors & Ventilation

- **Grid alignment:** The N-S/E-W grid can channel winds effectively when aligned with prevailing wind directions. Summer prevailing winds in Phoenix are generally from the south to west-southwest. E-W streets partially align with these winds, potentially facilitating some convective cooling. N-S streets act as cross-corridors.
- **Wind speeds:** Average summer wind speeds in Phoenix are relatively low (~6–8 mph), and the flat terrain and sprawling urban form offer little topographic channeling. Nighttime winds are often very calm (2–4 mph), severely limiting convective heat dissipation — a key factor in the intense nocturnal UHI.
- **Open urban fabric:** Paradoxically, Phoenix's wide streets and abundant surface parking lots allow substantial solar access to ground surfaces during the day, maximizing heat absorption. At night, the open geometry allows slightly better radiative cooling than deep canyons would, but the vast quantity of stored thermal energy in the extensive impervious surfaces overwhelms this advantage.

Sky View Factor (SVF)

- **Estimated SVF:** In most of downtown Phoenix, the SVF is relatively high — estimated at**0.5–0.8** across most streets, reflecting the wide streets and moderate building heights. In the tightest canyon configurations between the tallest buildings, SVF may drop to 0.3–0.4. In suburban residential areas (the vast majority of the metro), SVF approaches 0.85–0.95.
- **Nocturnal cooling implications:** High SVF generally facilitates faster nocturnal radiative cooling. Despite this, Phoenix's nighttime temperatures remain extremely elevated (July overnight lows rarely dropping below 85–90°F in the urban core). This demonstrates that the dominant UHI drivers in Phoenix are **thermal mass of extensive impervious surfaces, anthropogenic heat from AC waste heat rejection, and limited evaporative cooling** rather than geometric radiation trapping. The high SVF allows the stored heat to radiate toward the atmosphere but the sheer volume of stored energy keeps temperatures high.

Sources:

- Oke, T.R. — "Boundary Layer Climates" (urban canyon theory and SVF concepts)
- Arizona State University — Urban Climate Research Center (Phoenix UHI characterization)
- City of Phoenix street design standards
- Stewart and Oke — Local Climate Zone classification framework

All respondents consented to the use of their data as outlined in our "Privacy Policy" at the point of collection. Privacy Policy